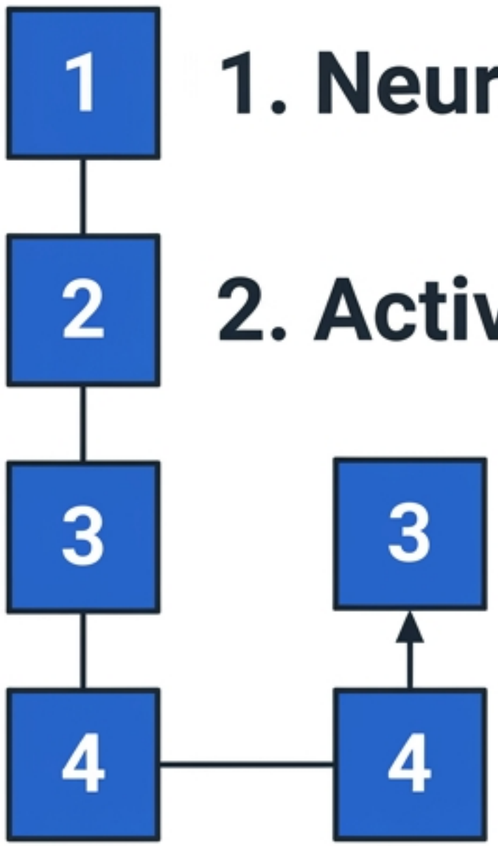
MODULE 11, LESSON 1

AI 101 for Mechanical Engineers

A Neuron is a Small Computation

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Course Outline



1. Neuron Mathematics
2. Activation Functions
3. Layers & Nonlinearity
4. Parameter Updating

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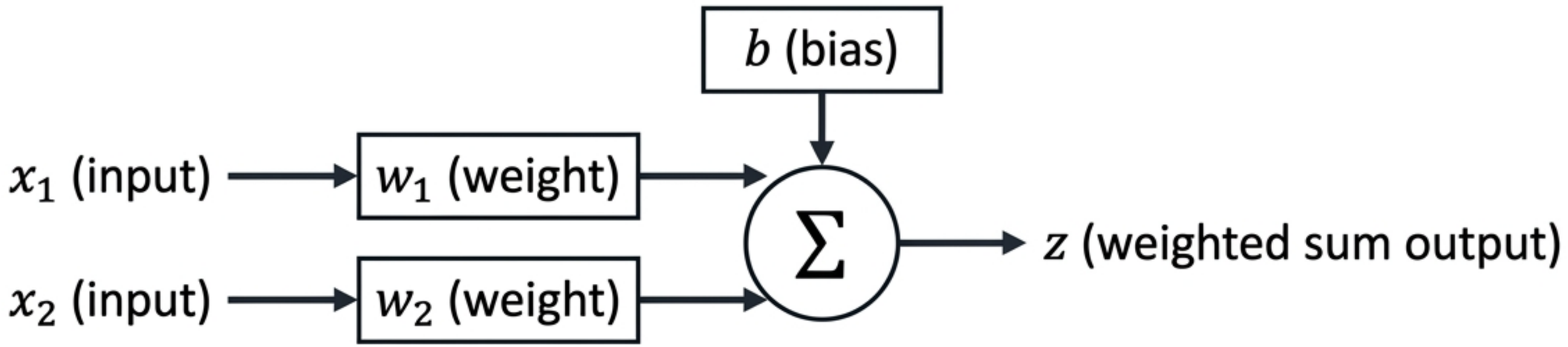
1

Neuron Mathematics

Define the affine transformation of a basic computational unit.

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The Artificial Neuron Model



- An artificial neuron computes an affine mathematical transformation.
- Weights (w_i) act as learned multipliers scaling input contributions.
- Bias (b) acts as an additive parameter shifting the weighted sum.
- The computation yields a weighted sum: $z = \sum w_i \, x_i + b$
- All variables and outputs are strictly dimensionless quantities.

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Vocabulary: **Parameters** vs. **Structure**

Fitted Parameters (Dynamic learned elements)	**Structural Elements** (Static computational architecture)
➔ **Weight:** A learned multiplier scaling a specific feature.	➔ **Neuron:** A distinct unit computing inputs, bias, and activation.
➔ **Bias Term:** An additive shift, distinct from statistical bias.	➔ **Layer:** A specific collection or stage of neural computations.
➔ **Fitted Connection Coefficients:** Synonymous with learned weights.	

- **Parameters** adjust the signal during the training process.
- **Structural elements** define the fixed routing of the model.

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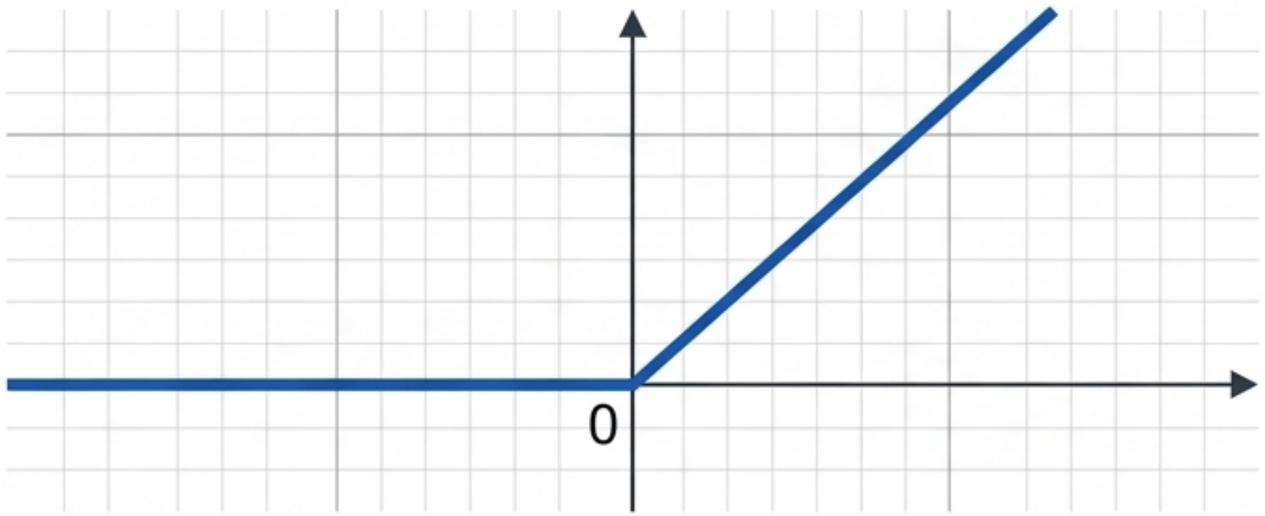
2

Activation Functions

Introduce the **Rectified Linear Unit** (ReLU) nonlinear filter.

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Rectified Linear Unit (ReLU)



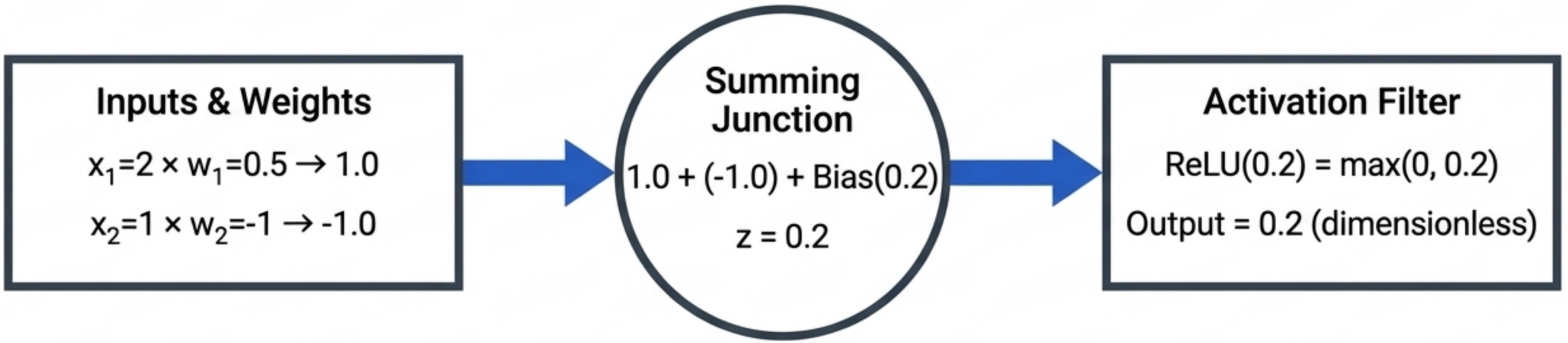
$$f(z) = \max(0, z)$$

(Units: dimensionless)

- Activation functions transform a neuron's weighted sum output.
- ReLU introduces critical nonlinearity into the signal path.
- Positive input values remain entirely unchanged.
- Negative input values are clipped strictly to zero.
- For example, an input of $z = -3$ yields 0.

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Feedforward Computation Example



- Multiply inputs by respective weights to find scaled contributions.
- Add scaled contributions and the bias to compute z.
- Pass z through the ReLU filter to compute activation.
- The final output value is mathematically deterministic.

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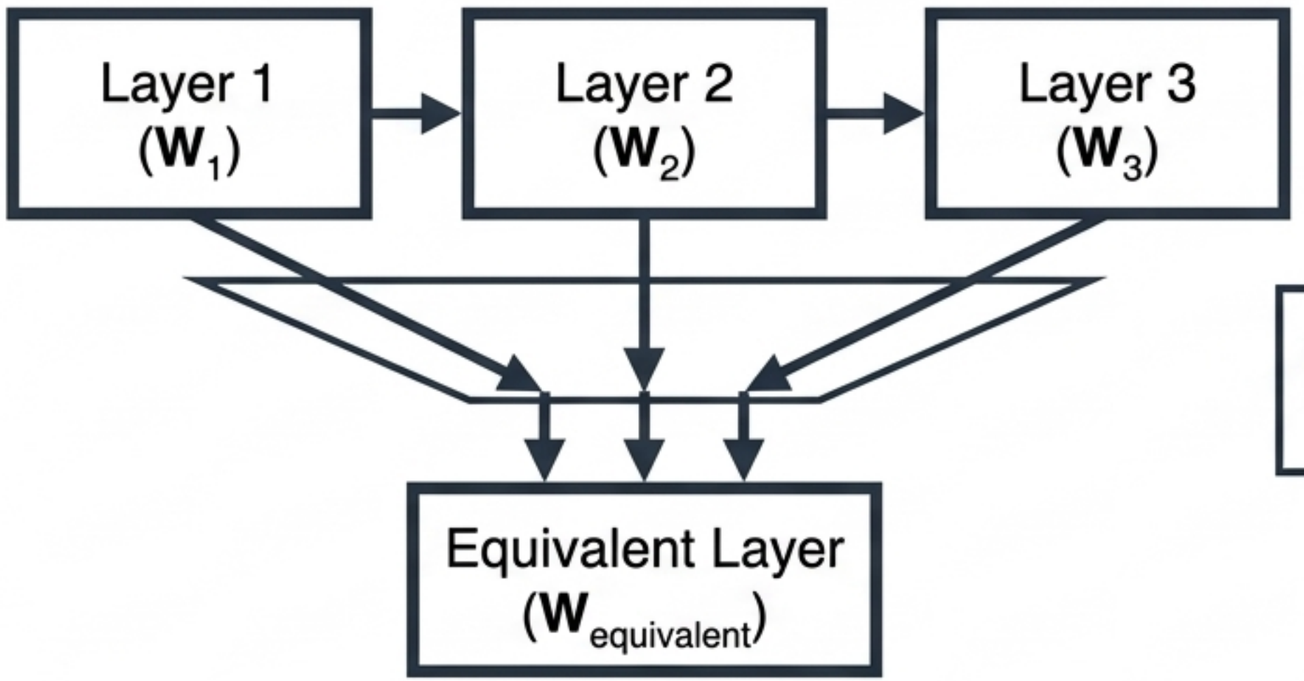
3

Layers & Nonlinearity

Prove the structural necessity of non-linear intersections in deep networks.

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Stacked Affine Mappings

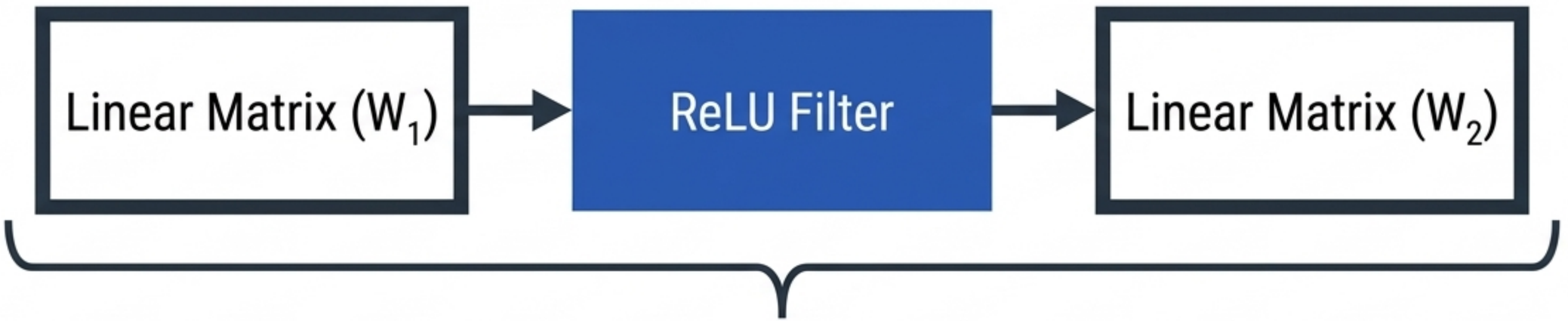


$$W_3(W_2(W_1\,x)) = W_{equivalent}\,x$$

- Networks combine multiple stages of layers into complex mappings.
- Composition of purely affine mappings remains mathematically affine.
- Stacked affine layers collapse to a single equivalent mapping.
- Adding layers alone cannot generate a useful non-linear model.

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The Role of Nonlinear Intersections



- Activation functions intercept signals between sequential linear layers.
- This interception prevents the mathematical collapse of deep architectures.
- Non-linearity allows networks to map complex physical phenomena.
- Deep architectures mandate non-linear operations to function effectively.

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4

Parameter Updating

Define the basic mechanics of parameter optimization via gradient descent.

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Gradient Descent Application

$$\theta_{new} = \theta_{old} - (learning_rate \times gradient)$$

$$1.95 = 2.0 - (0.1 \times 0.5)$$

- Backpropagation calculates objective gradients across the network architecture.
- The chosen optimizer applies a mathematical parameter update rule.
- Learning rate acts as a training setting, not a parameter.
- A single parameter update guarantees no global optimum convergence.

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Key Points

1. Neuron Mathematics: A neuron mathematically computes a dimensionless weighted sum alongside an additive bias.
2. Activation Functions: The ReLU function introduces nonlinearity by clipping negative numerical inputs to zero.
3. Layers & Nonlinearity: Nonlinear intersections prevent stacked affine layers from collapsing into single mappings.
4. Parameter Updating: Gradients dictate parameter adjustments, though single updates cannot guarantee global optimization.